The User Acceptance of Service Desk Application System Description

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***Abstract* – User acceptance is an important factor in the successful implementation of the system and applications in the context of higher education institutions. Since UIN Syarif Hidayatullah Jakarta currently has applied Service Desk Application System (SDAS) as a single point of contact to developer team where the user encounters the problem related to all IT services provided by the institution. UIN Syarif Hidayatullah Jakarta has decided to implement service desk application system (SDAS) that managed by The Centre of Information Technology (PUSTIPANDA). Since this system already implemented, many SDAS users did not take care to employ the application system. Many of users did not know the implementation of SDAS accordingly. Moreover, the negative perceived of the users have established right now, hence they prefer and more tend to use the manual system actually. This study attempt to describe the perception of user acceptance of SDAS using the Unified Theory of Acceptance and Use of Technology (UTAUT) model. Using the survey techniques, the study provides the description of the implementation of SDAS in the university.**

Keywords: User Acceptance, Service Desk Aplication, Help Desk, UTAUT

I. INTRODUCTION

Introduction the new application of information technology (IT) to the organization is not trivial process, since the application of technology will affect the whole organization, especially for human resources. IT implementation can be supposed to be successful when organizations that adopt such IT innovations have benefited from the application of IT to their organizations [1]. Venkatesh [2] states that the use of technology can increase the productivity of the organization but with certain conditions. The adoption of such technology must be accepted and used by members of the organization as internal users. The willingness of users in receiving technology is one of the key to the successful implementation of information technology innovation [20] [21]. Some research shows that the failure of the application of information technology today is more due to aspects of the behavior of information technology users.

User acceptance has been viewed as an important factor in determining the success or failure of an information systems project [3]. In the field of education, the use of IT also provides great benefits for various activities related to teaching and learning process, administration, as well as media introduction and promotion from the college to the community. State Islamic University (UIN) Syarif Hidayatullah Jakarta, as one of the largest Islamic universities in Indonesia that has a vision to become a world-class university and research university with the advantages of scientific integration, Islamicity and Indonesian, where IT/IS becomes a part it is important to realize that vision, especially for the Center for Information Technology (PUSTIPANDA). PUSTIPANDA as the technical unit that operationalize and manage related Information and Communication Technology (ICT) within UIN Jakarta.

This study aim to describe the perceive of user acceptance of SDAS in UIN Syarif Hidayatullah Jakarta using the model of UTAUT.

II. LITERATURE REVIEW

A. *User Acceptance*

Dillon & Morris [5] define the user acceptance as the willingness of a group of users to use information technology to support their work. Lack of user acceptance is a significant obstacle to the successful implementation of a new information system. Even users often do not want to use the information system that has been provided, but if the user is willing to use will generate profits for the user. There are several theories of user acceptance as in Table I.

TABLE I
Models and Theories of Individual Acceptance [4]

Models and Theories	**Constructs**
The Theory of Reasoned Action (TRA) proposed by Fishbein and Ajzen [6]. The theory to be used to measure behavioral intention and performance.	Attitude Subjective norm
The established theory of Technology Acceptance Model (TAM) by Davis [7]. The theory to determine user acceptance of technology. The next TAM 2 → Technology Acceptance Model 2 proposed by	Perceived Usefulness (PU) Perceived Ease of Use (PEU) Experience (TAM 2) Job Relevance (TAM 2) Subjective Norm (TAM 2) Voluntariness (TAM 2) Image (TAM 2)

Venkatesh and Davis [8]. This theory was adapted from TAM and incorporates additional variables.	Output Quality (TAM 2) Result Demonstrability (TAM 2)
Motivational Model (MM). This model shoots from psychology to explain behavior. Davis et al. [9] put on this model to the technology use and adoption.	Extrinsic Motivation Intrinsic Motivation
The Theory of Planned Behavior (TPB) proposed by Ajzen [10]. This theory covers TRA by incorporating additional variable to determine intention and behavior.	Attitude Perceived Behavioral Control Subjective norm
The model called by (C-TAM-TPB). This model combined TAM and TPB, proposed by Taylor and Todd [11].	Perceived Ease of Use Perceived Usefulness Attitude Perceived Behavioral Control Subjective norm
The Model of PC Utilization (MPCU) proposed by Thompson et al. [12]. This model was adjusted from the theory of attitudes and behavior Triandis [13]. It was used to predict PC usage behavior.	Affect Social Factors Perceived Consequences Habits Facilitating Conditions (Complexity, Job-Fit, Long-Term Consequences of Use)
The Diffusion of Innovation (DOI) theory proposed by Rogers [14]. This theory was adapted to information systems innovations by Moore and Benbasat [14].	Compatibility (DOI) Observability (DOI) Complexity (DOI) Relative Advantage (DOI) Trialability (DOI) Image Voluntariness of Use
Social Cognitive Theory (SCT) proposed by Bandura [15]. This theory was applied to information systems by Compeau and Higgins (1995). It is used to determine the usage.	Others' Use Affect Anxiety Support Self-Efficacy Encouragement by Others Performance Outcome Expectations Personal Outcome Expectations
The popular theory like Unified Theory of Acceptance and Use of Technology Model (UTAUT). This theory proposed by Venkatesh et al. [2]. This theory to measure user intention and usage on technology	Effort Expectancy Performance Expectancy Attitude toward Using Technology Social Influence Facilitating Conditions Self-Efficacy Anxiety

B. *Service Desk*

The service desk is the single point of contact where the company's internal staff will first contact this functional unit when they have questions or problems related to the application and IT. Requests made by the user to the service desk usually include access to information, problem-solving or knowledge sharing [19]. Service Desk is a functional unit consisting of a number of employees who are dedicated to handling various services. It is often done through telephone calls, web interfaces, or infrastructure events that are automatically reported by a system.

III. MODEL OF UNIFIED THEORY OF ACCEPTANCE AND USE OF TECHNOLOGY (UTAUT)

This study uses Unified theory of acceptance and use of technology (UTAUT) model to describe user acceptance [2] consisting of six variables, performance expectancy (PE), effort expectancy (EE), social influence (SI), facilitating conditions (FC), behavioral intention (BI) and use behavior (UB). Unified Theory of Acceptance and Use of Technology (UTAUT) is one of the accepted theories of technology developed by Venkatesh et al [2]. The UTAUT model is one of the most comprehensive, powerful, and up-to-date. UTAUT brings together eight pre-existing technology acceptance theories, namely the theory of reasoned action (TRA), the technology acceptance model (TAM), the motivation model (MM), the combination of TAM and TPB, the model of PC utilization (MPTU), innovation diffusion theory (IDT) and social cognitive theory (SCT) covering important elements of such models such as performance expectancy, effort expectancy, social influence, facilitating conditions, self-efficacy, attitude toward using technology, and anxiety. UTAUT has proven more successful in explaining up to 70 percent of intention variants when compared to the other eight theories in technology acceptance scenarios [5].

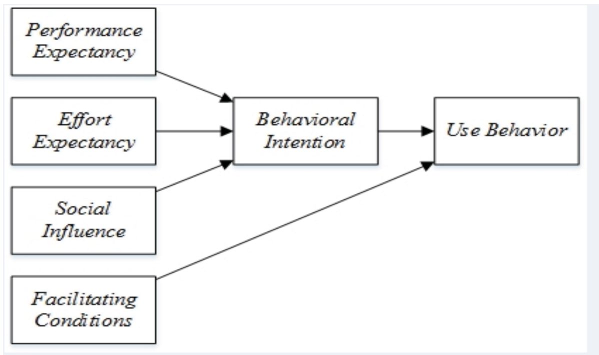

Figure 1. Model of Unified Theory of Acceptance and Use of Technology

IV. RESEARCH METHOD

This research using the quantitative method. Data collection was conducted through surveys by distributing questionnaires adapted from relevant previous studies [2] [4] [17]. The questionnaire was designed in the form of open questions based on variables and indicators on the UTAUT model. In addition, literature studies and interviews also conducted by researchers to strengthen the background and theories in this study.

The stratified random sampling method was used in this study, by determining the number of samples using the PLS-SEM approach [18].

SURVEY

Surveys were conducted by distributing questionnaires directly (paper based) and indirectly to SDAS users, i.e. students, staff and lecturers. Direct dissemination done by researchers to find the right responders through face to face directly. Indirect dissemination by researchers through social media application such as WhatsApp and line by using the google apps like google forms for distributing and responding.

Population and Sampling

The population in this study is divided into 3 groups of users namely, students, lecturers and employees. To determine the number of samples was done based on the criteria of the number of samples with the approach of Structural Equation Model (SEM), which is at least 10 times the number of paths in the research model [18], where in this study there are five paths (hypothesis) so the minimum sample is 50. SDAS user population consisting of 24,246 students, 571 employees, and 961 lecturers. However, because the number of each population is unbalanced where the number of employees and lecturers is very small compared to the student stratum, this is in line with data on the number of incoming complaints where complaints entering through SDAS are dominated by students with a percentage of 82.2% then used stratified random sampling technique (based on the user strata consisting of students, staff and lecturers).

Research Instrument

This study adapted the research instrument from UTAUT model to develop a questionnaire to describe the perceive of user acceptance and use of SDAS. The questionnaire consisted of six questions about the respondent's profile, 1 question about the experience of using IT services and two questions about the experience of using SDAS, and 21 test questions that have been adapted to the variables present in the UTAUT model. Furthermore, to ensure the validity and reliability of this questionnaire, researchers adopted a number of indicator items from a number of previous related studies [2] [4] [17].

V. THE RESULTS

This section presents findings from survey research that have been done by distributing questionnaires either directly (paper based) or online.

PROFIL OF RESPONDEN

In this study, there are questions related to the profile of respondents such as gender, age, employment status, work unit, recent education, formal education background in IT, knowledge of system existence, and experience in using the system.

Gender

Sex as shown in Table II comparison of respondents from this study men more with the percentage of 57% while women 43%.

TABLE II
JENIS KELAMIN RESPONDEN

Gender	Number	Percentage (%)
Male	81	57
Female	60	43
	141	100.0

AGE

Table III shows the characteristics of respondents based on their age.

TABLE III
AGE OF RESPONDEN

Characteristic	Number	Percentage (%)
17 – 25 years	128	91
26 – 35 years	3	2
36 – 45 years	4	3
More than 45 years' old	6	4
	141	100.0

From the results of this study obtained the results that the majority are in the age range from 17-25 years old. This study corresponds to the number of populations dominated by students who are in that age range.

STATUS OF THE RESPONDENT

Table IV shows the work status of respondents, respondents in this study are dominated by students as much as 128 people (91%), followed by nine as a lecturer (6%) and four of support staff (3%). This is in line with the data on the number of students with a total population of 24,246 people, while the staff and lecturers are 1530 people.

TABLE IV
STATUS OF RESPONDEN

Status	Number	Percentage (%)
Students	128	91
Support Staff	4	3
Lecturer	9	6
	141	100.0

WORKING UNIT OF RESPONDENT

Based on table V, it can be seen that the respondents in this study mostly came from Faculty of Science and Technology with 43 respondents (31%) continued from Faculty of Tarbiyah (education) as many as 19 respondents (14%). Respondents were the least obtained from the Rectorate and PUSTIPANDA employees with two persons. The majority of respondents are students dispersed throughout over whole faculty, hence the number of respondents from the faculty member are much more, while respondents from the Rectorate and PUSTIPANDA very small portion because it only consists of less of employees.

TABLE V
UNIT OF AFFILIATION

Unit	Number	Percentage (%)
FST	43	31
PUSTIPANDA	2	1
FISIP	11	8
FSH	9	6
FIDKOM	7	5
FEB	17	12
FAH	4	3
FITK	19	14
FDI	3	2
FKIK	13	9
F. Ushuluddin	7	5
F. Posology	4	3
Rectorate	2	1
	141	100.0

LEVEL OF EDUCATION

Table VI it can be seen that the respondents in this study mostly dominated by high school graduates as much as 128 respondents (91%), followed by graduate from master level S2 as many as 5 people (3%), then graduates from bachelor (S1) and the doctoral level (S3) with the same quantity in each category i.e. as many as 4 people (3%). This is because the majority of respondents in this study are active bachelor students who have the education level is junior high school (SMA).

TABLE VI
EDUCATION LEVEL

Category	Number	Percentage (%)
SMA	128	91
S1	4	3
S2	5	3
S3	4	3
	141	100.0

FORMAL EDUCATION BACKGROUND in IT FIELD

Table VII it can be seen that the amount of 86 respondents (61%) do not have educational background in information technology, and then the number of 55 respondents (39%) have formal education background in information technology field. This is in line with the number of respondents who come from the Faculty that there is no learning about information technology, and UIN Syarif Hidayatullah Jakarta itself is a State Islamic University dominated by the Faculty with Islamic education as its base.

TABLE VII
FORMAL EDUCATION BACKGROUND in IT

IT BACKGROUND	Number	Percentage (%)
Yes	55	61
No	86	39
	141	100.0

USER FAMILIARITY OF SDAS

Table VIII can be seen that the respondents in this study from the number of 141 respondents, the majority already know about the existence of service desk application system (SDAS) that is as much as 102 respondents (72%), the rest as many as 39 people (28%) do not know about the existence of SDAS. This shows that actually SDAS is well known by the users.

TABLE VIII
FAMILIARITY OF THE USER WITH SDAS

Familiarity	Number	Percentage (%)
Yes	102	72
No	39	28
	141	100.0

THE EXPERIENCE USING OF SDAS

Table IX it can be seen that respondents in this study as many as 88 respondents (62%) have used the service desk application system (SDAS), while the remaining 53 respondents (38%) have never been used SDAS. This is because SDAS is an application that is only used when users have problems related to IT services in UIN Syarif Hidayatullah Jakarta, so only those who have problems or request certain services to the PUSTIPANDA only who use SDAS.

TABLE IX
USER'S EXPERIENCE USING OF SDAS

Experiences	Number	Percentage (%)
Yes	88	62
No	53	38
	141	100.0

VI. CONCLUSION

Based on the results of the study indicated that the amount of 141 respondents who participated consisted of students, staff, and lecturers at UIN Syarif Hidayatullah Jakarta. The profile of respondent described the domination of the students with the age range of 17-25 years old with the level of education is senior high school. Furthermore, 102 respondents (72%) already know the existence of SDAS and 53 respondents (38%) the rest do not know yet. Of these, 88 respondents (62%) have used SDAS. This indicates that the respondents of this research have known the existence of SDAS and use it to support in terms of reporting problems related to IT services in UIN Syarif Hidayatullah Jakarta.

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